

# Algebra I Quadratics Cheatsheet

## Module 9

### Forms

Standard form:

$$y = ax^2 + bx + c$$

Factored form:

$$y = a(x - r_1)(x - r_2)$$

Vertex form:

$$y = a(x - h)^2 + k$$

### Graph Features

Vertex:

$$(h, k) \text{ from vertex form}$$

Axis of symmetry:

$$x = h$$

Opening:

$$a > 0 \Rightarrow \text{opens up}, \quad a < 0 \Rightarrow \text{opens down}$$

$$|a| > 1 \Rightarrow \text{narrower}, \quad 0 < |a| < 1 \Rightarrow \text{wider}$$

### Factoring to Solve

$$ax^2 + bx + c = 0$$

Factor and use zero product property:

$$AB = 0 \Rightarrow A = 0 \text{ or } B = 0$$

### Pure Quadratics

$$ax^2 + c = 0$$

Isolate  $x^2$ , then square root both sides:

$$x^2 = 25 \Rightarrow x = \pm 5$$

### Quadratic Formula

$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## Discriminant

$$D = b^2 - 4ac$$

$D > 0$  two real solutions

$D = 0$  one real solution

$D < 0$  no real solutions

## Intercepts

$y$ -intercept:  $(0, c)$

$x$ -intercepts: solve  $ax^2 + bx + c = 0$

## Quick Checks

- Set the quadratic equal to zero before factoring to solve.
- Include both signs when taking square roots.
- Vertex gives maximum if  $a < 0$  and minimum if  $a > 0$ .
- Use exact radical form unless decimals are requested.